

AHMAD MARAQA

AI / ML Engineer | AI Research & Integration Engineer

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[LinkedIn](#) · [GitHub](#) · [Portfolio \(link to be added\)](#)

PROFILE

AI/ML Engineer with professional experience developing AI-powered applications, intelligent automation workflows, machine learning solutions, and LLM-based systems. Hands-on experience with Retrieval-Augmented Generation (RAG), embeddings, vector databases, prompt engineering, local and open-source LLMs, and computer vision. Experienced integrating AI systems with databases, APIs, messaging platforms, and real-world IoT/RTLS environments using Python and n8n.

PROFESSIONAL EXPERIENCE

AI Research & Integration Engineer — *Penguin Location Services* Jan 2025 – Present · Amman, Jordan

- Research and implement AI/ML solutions for real-world applications, including machine learning models for IoT and RTLS environments.
- Work with BLE and IoT devices and controllers in RTLS applications, applying machine learning models.
- Develop and experiment with LLM-based applications using Retrieval-Augmented Generation (RAG), embeddings, vector databases, and local/open-source LLMs.
- Apply prompt engineering techniques when developing and improving LLM-based applications.
- Use Ollama for working with local LLMs and AI application development.
- Build AI-powered assistants and integrations connecting language models with external services and tools.
- Develop and maintain production automation workflows using n8n, MongoDB, APIs, webhooks, and external integrations.
- Integrate WhatsApp Business API into automated workflows and AI-powered applications.

AI Research Engineer Intern — *Penguin Location Services* Oct 2024 – Jan 2025 · Amman, Jordan

- Worked with BLE and IoT devices in RTLS applications.
- Trained and evaluated machine learning models including KNN, Decision Trees, and Random Forest.
- Implemented a custom probabilistic model for an RTLS-related application.

AI & ML Engineer Intern — *Leading Point Software* Jul 2024 – Sep 2024 · Amman, Jordan

- Worked on data science, machine learning, deep learning, and computer vision applications.
- Developed machine learning and computer vision solutions including image classification and object detection.
- Gained hands-on experience with AutoML tools.

SELECTED AI PROJECTS

AI Video Dubbing System — *Dublejli*

- Developed an AI-driven video dubbing system integrating Speech-to-Text (STT), text translation, and Text-to-Speech (TTS).
- Designed a synchronization algorithm to align generated speech with the original video.
- Exhibited at SOFEX Exhibition and Conference 2024 and the National Technology Parade (NTP) 2024.

Jordanian Currency Detection & Counting

- Developed a computer vision system for detecting and counting Jordanian currency using YOLOv8.

AI-Enhanced Smoke Detection

- Developed and evaluated a machine learning/computer vision solution for smoke detection using a Kaggle dataset.

TECHNICAL SKILLS

AI & LLMs: LLMs, Retrieval-Augmented Generation (RAG), Embeddings, Vector Databases, FAISS, Prompt Engineering, Agentic Workflows, AI Automation, Local & Open-Source LLMs

AI Tools & Frameworks: Ollama, LangChain, Langflow, Claude Code

Machine Learning & Computer Vision: Machine Learning, Deep Learning, Neural Networks, Computer Vision, Object Detection, Image Classification, OpenCV, TensorFlow, Keras, Ultralytics, CVAT, Data Annotation

AI Integration & Automation: Python, n8n, MongoDB, WhatsApp Business API Integration, APIs, Webhooks

Engineering Foundation: Git, GitHub, IoT, BLE, RTLS, Controllers, Verilog

EDUCATION

B.Sc. Computer Engineering — *University of Jordan*

Oct 2020 – Sep 2024 · Amman, Jordan

GPA: 3.77 / 4.00 — Excellent

Languages: Arabic, English